RUIZE JIN

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EDUCATION

DUKE UNIVERSITY, THE FUQUA SCHOOL OF BUSINESS — Durham, NC, USA
Master of Science in Quantitative Management: Business Analytics — *May 2027*
Coursework: Data Science for Business, Decision Analytics & Modeling, Modern Analytics, Data Infrastructure, Data Visualization
Teaching Assistant: Foundations of Business Analytics; Admissions Ambassador; Head of Outreach, Duke AI Competition Club

CENTRAL UNIVERSITY OF FINANCE AND ECONOMICS (CUFE) — Beijing, China
Bachelor of Science, Actuarial Science — *Jun 2026*
Student Ambassador, IFoA; Vice President, CUFE News Agency; Deputy Editor, University Newspaper; Stand-Up Comedian

EXPERIENCE

SCOR SE - ***Marketing Actuary Intern*** — *Beijing, China* — *Nov 2025 - Apr 2026*
- Developed a SQL ETL pipeline for millions of claims and policy records across 20 clients, shortening processing time by 45% and saving $10K+ for solvency analysis.
- Created Power BI dashboards for claims experience and performance analysis across 400+ insurance portfolios, incorporating quarterly data refreshes.
- Automated data extraction and validation from insurers' solvency reports with an AI agent, cutting report-processing time by 70%.

SIEMENS - ***Data Analytics Intern*** — *Beijing, China* — *Jul 2025 - Nov 2025*
- Automated economic data collection with Selenium and AWS pipelines; optimized AWS Glue Spark workflows, improving resource utilization by 30% and reducing processing time by 45%.
- Developed scripts to scrape, structure, and ingest 3,000+ business documents into a RAG knowledge base for Siemens' in-house LLM chatbot.
- Cleaned and migrated COPA data in pandas and delivered a Tableau CPD dashboard to analyze sales performance and inform order growth initiatives.

PEKING UNIVERSITY INTERNATIONAL HOSPITAL - ***Medical Data Operation Intern*** — *Beijing, China* — *Jun 2024 - Jul 2024*
- Reconciled quarterly inpatient data across 2,000+ records with reimbursement and finance teams, lowering errors by 32%.
- Presented DRG/DIP reimbursement analyses to visiting faculty and the dean, explaining financial implications for hospitals, patients, and insurers.

SELECTED PROJECTS

STEAM PLAYER ENGAGEMENT MODELING | Python, LightGBM, NLP, Snowflake — *Oct 2026 – Nov 2026*
Coordinated task allocation and progress tracking for a five-member project team. Modeled observed post-review engagement with LightGBM on 10.8M negative Steam reviews, combining behavioral features and 35 NLP-derived complaint factors. Compared four specifications with 10-fold cross-validation; reached 0.824 ROC-AUC on a 2.15M-review 2023 test set. Analyzed complaint patterns across four player profiles and implemented the workflow in Snowflake for ongoing analysis.

INTERPRETABLE MULTI-TASK INSURANCE PRICING | PyTorch, R, Deep Learning — *Sep 2025 - Apr 2026*
Implemented a PyTorch multi-task neural additive model (MTL-NAM) with shared feature subnetworks to jointly predict claim frequency and severity on freMTPL. Benchmarked against GLM, GBM, and single-task NAM with six-fold cross-validation, yielding severity MAE 4.4% below GBM while retaining feature-level interpretability for insurance pricing.

NYC TAXI TRIP PREDICTION | Python, XGBoost, FastAPI, Google Cloud — *Mar 2025 - Jul 2025*
Benchmarked CNN, Attention-LSTM, and XGBoost; trained XGBoost models on 15M+ NYC taxi trips with 27 pre-trip features. Achieved 4.01-minute duration MAE and $3.66 pre-tip cost MAE on a held-out future-month test. Deployed an interactive prediction application with FastAPI, Docker, and Google Cloud Run, mapping user-selected locations to taxi-zone features.

PROMPT2PRINT: GENERATIVE AI WORKFLOW | Agent, FastAPI, React — *Sep 2026*
Built an AI workflow generating four typography proposals and converting selected designs into 3D-printable STL files. Integrated 15 geometry checks and up to two AI repair iterations; evaluated pipeline reliability, increasing yield from 81.9% to 97.5% and saving an average of 23 minutes per candidate.

TECHNICAL CAPABILITIES

Programming: Python (pandas, NumPy, PySpark, PyTorch, Transformers), SQL, R
Platforms: AWS (AI Practitioner Certification), Snowflake, Google Cloud, Databricks, Docker, FastAPI, Hugging Face, Git
Modeling: XGBoost, LightGBM, Deep Learning, Cross-Validation, Multi-Task Learning, NLP, RAG, LLM Agents, Fine-Tuning

ADDITIONAL INFORMATION

Cooking (American & Chinese BBQ), Bird Lover, 3D Printing, Stand-Up Comedy